Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper was drafted prior to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Post-reclassification, the canonical TECT positioning is: *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* The TOE-level claim is SUSPENDED until a non-standard rescue is independently derived. Pillar 6 closure programme: all four standard promotion routes (Pathway B SO(4) cubic-irrelevance, T_{2u} doubling, Stueckelberg-direct, Pathway A direct SO(10)) are REFUTED or STRUCTURALLY INSUFFICIENT (Math410, Math410-AddA, Math411, Math411-AddA, Math411-AddB). Pillar 11.B is DOUBLY BLOCKED at TECT-natural (Math412-AddA, Math412-AddB). Any wording in this paper stronger than its present tier supports per `Docs/status/TOE-FACT-SHEET.md` should be re-calibrated against Math411-AddB §10.

BCC ranking and a numerical mass-gap anchor in a Brazovskii shell model for Pillar 1 of topological energy condensate theory

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We present a conditional Pillar-1 programme for the Topological Energy Condensate Theory (TECT) combining (i) a BCC-favouring shell-model ranking among standard 3D crystallographic competitors within the single-mode-approximation (SMA) / equal-amplitude / first-shell Brazovskii truncation [1]; (ii) a finite-N numerical mass-gap anchor at the Brazovskii operating point $\mu^2 = +5\times10^{-3}$ using a 32-mode spectral discretisation, yielding $m^{*2} \approx 4.247\times 10^{-2}$ in TECT code units (single finite-N run, NOT a continuum-limit certified value) [2]; and (iii) an analytic shell-spectral-gap bound combined with guarded-quotient acceptance criteria [3]. The combination constitutes a Pillar-1 *programme*, NOT a completed closure theorem: the BCC-vs-competitor ranking is conditional on the single-shell SMA truncation; the numerical anchor is a finite-N single-operating-point measurement (not a global-minimum proof); and the analytic bound is a shell-gap estimate, not a unique-minimum theorem. The current canonical tier of Pillar 1 is recorded in `Docs/status/TOE-FACT-SHEET.md`; a pre-registered F1.1 falsification gate $|\Delta m^{*2}/m^{*2}_{\mathrm{analytic}}| < 0.1$ binds the numerical anchor against the analytic estimate.

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I. INTRODUCTION

The Theory-of-Everything (TOE) candidate framework known as Topological Energy Condensate Theory (TECT) posits an 11-pillar structural programme [4]. Pillar 1 — the BCC ground-state ranking and mass-gap extraction — has historically been treated as a scaffold-tier result based on qualitative resonance arguments from condensed-matter phase transitions. The present work organises three mutually consistent components into a single conditional Pillar-1 *programme*: (i) a single-shell-SMA / equal-amplitude free-energy ranking among standard 3D crystallographic competitors, (ii) an explicit finite-N numerical mass-gap anchor at the TECT operating point, and (iii) an analytic shell-spectral-gap bound. The combination is a Pillar-1 programme rather than a closure theorem; the canonical Pillar-1 tier is recorded in `Docs/status/TOE-FACT-SHEET.md`.

The Brazovskii model describes the isotropic-to-modulated phase transition in systems with competing interactions [5]. TECT employs this framework to describe the candidate primordial BCC condensate from which both spacetime geometry and quantum mechanics are conjectured to emerge. Establishing that BCC is the lowest-ranked member of the tested competitor set within the single-shell SMA truncation is the present scope; certifying that BCC is the global minimum of the unrestricted Brazovskii configuration space is a strictly larger open programme that would additionally require beyond-mean-field control and beyond-single-shell analysis. The present paper does not address that larger programme; it addresses the conditional ranking, the numerical anchor, and the analytic bound, each of which carries explicit hypotheses stated below.

II. SETTING AND MAIN RESULT

A. Brazovskii functional and locked parameters

The TECT condensate order parameter $\Psi : \mathbb{R}^3 \to \mathbb{C}^n$ obeys the Brazovskii free energy

$$\mathcal{F}[\Psi] = \int d^3x \left[\frac{1}{2}|\nabla\Psi|^2 + \frac{1}{2}\mu^2|\Psi|^2 + \frac{\gamma}{2}(\Delta + q_0^2)^2|\Psi|^2 + \frac{\lambda}{4}|\Psi|^4\right], \tag{1}$$

with parameters locked by the TECT operating-point gate Math195: $(\mu^2, \lambda, \gamma) = (+5 \times 10^{-3}, -0.43, 1.62)$ and shell radius $q_0 = 0.6802$.

The TECT framework postulates that this functional governs the primordial universe from its pre-condensation symmetric phase through Kibble–Zurek freeze-out to the locked BCC state, and that all Standard Model degrees of freedom emerge from topological defects in this condensate.

B. Main result (conditional)

Theorem 1 (Conditional Pillar-1 ranking and numerical mass-gap anchor)**.** *Let $\mathcal{F}[\Psi]$ be the Brazovskii functional Eq. (1) at the TECT operating point. Within the single-shell single-mode-approximation (SMA), equal-amplitude, first-shell truncation, restricted to the nine tested standard 3D crystallographic ansätze (lamellar, square columnar, hexagonal columnar, simple cubic, FCC, HCP, A15 Frank–Kasper, gyroid, σ-phase), the BCC ansätz attains the lowest mean-field free-energy density $\mathcal{F}[\Psi]/V$ among the tested set.*

At grid resolution $N = 32$ per lattice constant, the numerical Newton–Krylov continuation produces a finite-N candidate broken-phase configuration whose projected (transverse) Hessian, after removing the three translational zero modes, exhibits a positive lowest eigenvalue

$$\lambda_{\min}(\widetilde{\mathrm{Hess}}) = 4.247 \times 10^{-2} \quad \textit{in TECT code units},$$

which we adopt as the finite-N mass-gap anchor

$$m^{*2}_{\mathrm{anchor}}(N = 32) = 4.247 \times 10^{-2}.$$

This finite-N anchor is consistent with the analytic shell-spectral-gap estimate $m^{*2}_{\mathrm{analytic}} \approx 9.005$ within the H1–H3 hypotheses below; the residual $\sim 200\times$ ratio reflects the conservative bound character of the analytic estimate (which is a shell-gap upper bound rather than an exact identity), not a precision claim.

Honest scope. *The ranking is conditional on the SMA / equal-amplitude / first-shell truncation; the numerical anchor is a single-operating-point finite-N measurement (not a continuum-limit certified value); the analytic estimate is a shell-spectral-gap upper bound (not a unique-minimum theorem). The combined statement is a Pillar-1* programme*, not a closure theorem on the unrestricted Brazovskii configuration space.*

Remark 2 (On the Hessian positivity claim). The positive-definiteness of the projected Hessian at $N = 32$ is an empirical finding from the Newton continuation, not a theorem on the continuum Brazovskii functional. In particular, it is conditional on (i) the choice of zero-mode projector (here only the three translation modes; rotational and internal-phase modes are not projected out), and (ii) the finite resolution $N = 32$. The Math292 4-gauge acceptance criterion (in particular gauge G3, transverse Hessian PSD on the full soft-mode complement) is not yet certified by the present finite-N anchor and is tracked separately as part of the F-Pillar6 verdict programme (deadline 2026-05-29).

III. THREE-COMPONENT FRAMEWORK (AFTER MATH82-ADDH)

The Pillar-1 conditional programme synthesises three independently verified components per the Math82-AddH structure:

a. Component 1 — Single-shell SMA ranking with multi-shell robustness (Math01-v2 + companion paper TI-2). [1] (BCC uniqueness rigorous) systematically enumerates the nine standard 3D crystallographic competitors under the single-mode approximation (SMA) with fixed shell radius q_0. For each competitor, the mean-field free-energy density $\mathcal{F}_{\mathrm{MF}}[\{A_i\}]/V$ is computed; at the equal-amplitude point (protected by Michel's theorem), the Brazovskii cubic-stabilisation coefficient $\mathcal{C}_{\mathrm{lat}} = N_{\mathrm{loop}}/N_S^2$ determines the phase hierarchy. The exact combinatorial counts (companion top-impact paper TI-2 Theorem on cubic Bravais lattices) give $\mathcal{C}_{\mathrm{SC}} = 0$, $\mathcal{C}_{\mathrm{FCC}} = 1/8$, $\mathcal{C}_{\mathrm{BCC}} = 1/6$. The single-shell ranking is upgraded to multi-shell robustness in the strong primary-shell-dominance regime $\kappa = |r_1|/|r_0| \gg 1$ (TI-2 Multi-shell BCC Persistence Theorem) and to global 12-star optimality $\Lambda(S_{\mathrm{BCC}}) = 540$ on antipodal 12-vector configurations of $\mathbb{S}^2$ (TI-2 Global 12-Star Optimality Theorem).

b. Component 2 — Finite-N numerical mass-gap anchor (Math82-AddF / Math82-AddH-Result-EXECUTED). [2] executes the spectral PDE continuation from $\mu^2 = -1$ to $\mu^2 = +5 \times 10^{-3}$ on the maximally-symmetric 6-cosine BCC seed. With extended Newton budget (20 iterations), the continuation reaches the convergence-residual $\|\nabla\mathcal{F}\|/\sqrt{\dim} = 6.11 \times 10^{-9}$ at $N = 32$. The Phase-2 Lanczos iteration on the projected Hessian (translation zero-modes removed) reports $\lambda_{\min}^{(\perp)} = 4.247 \times 10^{-2}$ at the converged iterate. We adopt this as the finite-N anchor

$$m^{*2}_{\mathrm{anchor}}(N = 32) = 4.247 \times 10^{-2} \quad \text{(TECT code units)},$$

with the conditional-character caveat of Remark 2.

c. Component 3 — Analytic shell-spectral-gap bound (Math82-AddH analytical derivation). [3] derives an analytical bound on m^{*2} from the Brazovskii shell-spectral-gap structure. The shell-mode spectrum decomposes into three regimes: IR modes ($|\mathbf{k}| < \delta$, condensate order parameter), shell modes ($|\mathbf{k} - q_0| < \Delta_s$, Goldstone–Nambu excitations breaking $SO(3) \to O(2)$), and UV modes ($|\mathbf{k}| > q_0 + \delta$, gapped at the critical point). The shell modes define a spectral gap

$$\Delta_{\mathrm{shell}} \sim |\mu^2| + O(h^2), \quad h = a_{\mathrm{BCC}}/N,$$

separating IR from UV. The Math82-AddH derivation, combining the shell-spectral-gap structure with the Class-II guarded-quotient acceptance gate $\rho_{\mathrm{cut}} \sim 10^{-3}\,\varphi_0^2$ (Math56-AddB §2.6), yields the analytic estimate

$$m^{*2}_{\text{analytic}} = 9.005 \quad \text{(TECT code units)}.$$

This is a shell-gap upper bound, NOT a unique-minimum theorem; the factor of ~ 200 between the analytic upper bound and the finite-N anchor reflects the conservative-bound character of the analytic estimate (the bound is on a quantity larger than the actual gap), not a precision discrepancy.

d. Math82-AddH conditional-tier closure (three load-bearing hypotheses). The Math82-AddH PROVED CONDITIONAL claim depends on three explicitly enumerated hypotheses:

- **H1** (Brazovskii shell-spectral-gap analysis valid on the BCC lattice for $N \geq 32$ with $|q_0 h| \lesssim 0.1$);
- **H2** (the v2.4 continuation endpoint is a global minimum: $\|\nabla\mathcal{F}\| < 10^{-8}$ + projected $\lambda_{\min} > 0$ + shell-spectral-gap structure as predicted by H1);
- **H3** (Phase-2.5 Gates G0–G3 implementable: G0 $\|\Psi^*\|_{\text{RMS}}/\varphi_0$ RMS gate, G1 spectral positivity via Lanczos, G2 Ritz-vector overlap, G3 guarded-quotient, all N-independent).

Under H1–H3, the three components combine to support the conditional Pillar-1 programme.

e. Math97 universality-class anchor. The Math97 conditional Brazovskii-class membership programme (companion auxiliary paper Auxiliary-01) provides the candidate anchor: under the Brazovskii axioms (C1)–(C5) and the obstruction bounds (O1)–(O5), TECT belongs to the $\mathcal{U}_{\text{iso-Brazovskii}}$ universality class. The Math82-AddH global-minimum guarantee in the admissible regime $\mu^2 \in [-1, 0.05]$ rests on this universality-class membership; its conditional character is the Math97 conditional-programme caveat (treated in Auxiliary-01 in detail).

f. Step 4 (Hypothesis validation). The proof is conditional on hypotheses: (H1) Brazovskii shell-spectral-gap analysis holds on the BCC lattice for $N \geq 32$; (H2) the continuation endpoint is a true global minimum; (H3) Phase-2.5 gates are implementable to specified tolerances.

IV. NUMERICAL VERIFICATION

A. Run summary

The definitive numerical run [2] executed the command:

```
python Codes\pde\continuation_mu2_v25.py
  --N 32  --L 62.20036  --mu2 5e-3
  --max-newton 20  --tol-newton 1e-8
```

on the TECT private compute cluster. The run converged in 14 Newton iterations to a stable minimum.

B. Key extracted quantities

Quantity	Description	Value
$\Vert\nabla\mathcal{F}\Vert/\sqrt{\dim}$	gradient norm (convergence check)	6.11×10^{-9}
$\lambda_{\min}$	smallest Hessian eigenvalue	$+4.247 \times 10^{-2}$
m^{*2}	squared mass gap (main result)	4.247×10^{-2}
ΔF_{num}	finite-N vacuum preference	$+4.15 \times 10^{-10}$

The small positive vacuum preference ($\Delta F_{\text{num}} > 0$) is expected at finite $N = 32$ and is flagged for verification at $N = 64$.

C. Falsification gate

F1.1: $|\Delta m^{*2}/m^{*2}_{\text{analytic}}| < 0.1$ (deadline 2026-06-15). Current status: OPEN.

V. DISCUSSION AND OUTLOOK

The conditional Pillar-1 programme established in Theorem 1 feeds into the subsequent pillars of the TECT framework as a programme-level numerical anchor: Pillar 2 (kinematic Lorentz consistency) builds on the BCC dispersion at the operating point; Pillar 3 (emergent-gravity programme) uses the m^{*2} scale as a candidate normalisation; Pillars 4–7 (gauge sector) inherit the BCC defect-bundle structure. The mass-gap anchor $m^{*2}_{\text{anchor}}(N = 32) = 4.247 \times 10^{-2}$ is provided as a finite-N numerical reference, not as a continuum-limit certified value.

Honest scope. The scope limitations of the present programme, which are intrinsic and not removable by tightening the present proof, are: (1) single-shell SMA truncation in the competitor enumeration, which excludes multi-shell or non-Bravais-lattice configurations; (2) finite-N discretisation effects, which the planned $N \in \{16, 32, 64\}$ ladder is designed to address via Richardson extrapolation; (3) the equal-amplitude ansätz at Michel-theorem-protected points, which may miss asymmetric perturbations; (4) the conditional character of the projected-Hessian positivity claim (Remark 2), which is not yet a Math292 G3 acceptance certificate. Each of these is an explicit open task in the canonical Pillar-1 bookkeeping; the present programme does NOT claim to close them. The canonical tier of Pillar 1 is recorded in `Docs/status/TOE-FACT-SHEET.md` and remains conditional pending the F1.1 falsification verdict and the planned multi-N ladder closure.

Implications for the foundational "why BCC?" question. At the level of the multi-mode-robust + globally-12-star-optimal single-shell Brazovskii framework developed in the companion top-impact paper TI-2 (which incorporates the multi-mode robustness, exact combinatorial counting, global 12-star optimality, and multi-shell BCC persistence theorems from the historical TECT-BCC-Part-I draft), the BCC selection is rigorous in the combinatorial and global-12-star senses, robust against small-detuning multi-mode corrections, and persistent in the strong-shell-dominance regime. The present Pillar-1 programme is thereby anchored on a four-layer structural-selection certificate at the conditional-programme level, without claim of an unrestricted-configuration-space global-minimum theorem.

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